ADVANCED MACHINE LEARNING

ASSIGNMENT 3- TIME SERIES DATA

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1. Introduction

Weather forecasting is an important application of deep learning because weather conditions change continuously over time. Predicting future temperatures requires a model that can recognize patterns from previous observations and use them to estimate upcoming values. Recurrent Neural Networks (RNNs) are useful for this type of problem because they are designed to work with sequential data.

In this assignment, I explored different deep learning models to predict future temperature using historical weather measurements. My main goal was to understand how changes in the network architecture affect forecasting performance. I compared a GRU model, an LSTM model, a stacked GRU model, and a hybrid model combining one-dimensional convolutional layers with GRU. I also used a simple baseline approach to compare the deep learning models against a basic forecasting method.

For this experiment, I used the Jena Climate dataset, which contains weather observations collected over several years. I trained and evaluated the models using the same data preparation approach to make the comparison consistent. I compared their validation performance, tested the strongest models on unseen data, and examined the results to identify which architecture performed best for temperature forecasting.

2. Dataset Description and Data Preparation

For this assignment, I used the Jena Climate dataset, which contains weather measurements recorded between 2009 and 2016. The dataset includes 420,551 observations and 15 columns, with information such as temperature, atmospheric pressure, humidity, and other weather-related measurements. The main objective was to predict future temperature based on patterns observed in previous weather conditions.

Before training the models, I prepared the dataset to make it suitable for time-series forecasting. I divided the observations chronologically into three groups: 50% for training, 25% for validation, and 25% for testing. Keeping the original time order was important because weather forecasting depends on past observations, and randomly mixing the data could introduce information from the future into the training process.

I also normalized the weather measurements using statistics calculated from the training portion of the dataset. The same normalization was applied to the validation and test data to maintain a fair comparison and avoid data leakage.

For each prediction, the models used 120 hours of historical weather information to forecast the temperature 24 hours ahead. I followed the same data preparation and evaluation process for all the models so that differences in performance could be compared consistently.

3. Models Used in the Experiment

To understand how different neural network architectures affect weather forecasting, I experimented with four deep learning models. I also included a simple baseline method to compare their performance. Each model was trained using the same prepared weather dataset so that I could evaluate how the architecture influenced the prediction results.

GRU Model: I first used a Gated Recurrent Unit (GRU) model to learn patterns from historical weather observations. GRU networks are designed to process sequential information while maintaining important details from previous time steps. This model provided a starting point for comparing the performance of different recurrent architectures.

LSTM Model: Next, I implemented a Long Short-Term Memory (LSTM) model. LSTM networks use memory cells and gates to manage information over longer sequences. I wanted to examine whether this architecture could capture temperature patterns more effectively than the GRU model and produce more accurate forecasts.

Stacked GRU Model: I also experimented with a stacked GRU architecture containing multiple recurrent layers. The purpose of this model was to investigate whether increasing the depth of the network could help it learn more complex relationships in the weather data. I compared its performance with the simpler GRU and LSTM models to determine whether additional recurrent layers improved forecasting accuracy.

Conv1D + GRU Model: Finally, I developed a hybrid model combining one-dimensional convolutional layers with a GRU network. The convolutional component was used to identify local patterns in the weather measurements, while the GRU component processed sequential relationships. This approach allowed me to explore whether combining different deep learning layers could improve temperature predictions.

Baseline Method: Along with the neural network models, I evaluated a simple forecasting baseline that used the most recent observed temperature as the prediction. This provided a reference point for determining whether the more complex models offered an improvement over a basic forecasting approach.

4. Model Performance and Results

After training the models, I compared their performance using validation Mean Absolute Error (MAE). MAE measures the average difference between predicted and actual values, so a lower MAE indicates better forecasting performance.

The validation results from my experiment are summarized below.

Model	Validation MAE (Normalized)
GRU	**0.2704**
LSTM	**0.2637**
Stacked GRU	**0.2741**
Conv1D + GRU	**0.2671**

The last-observation baseline achieved a normalized validation MAE of 0.2783. All four neural network models achieved lower validation errors than this baseline, showing that they learned useful patterns beyond simply repeating the most recent temperature.

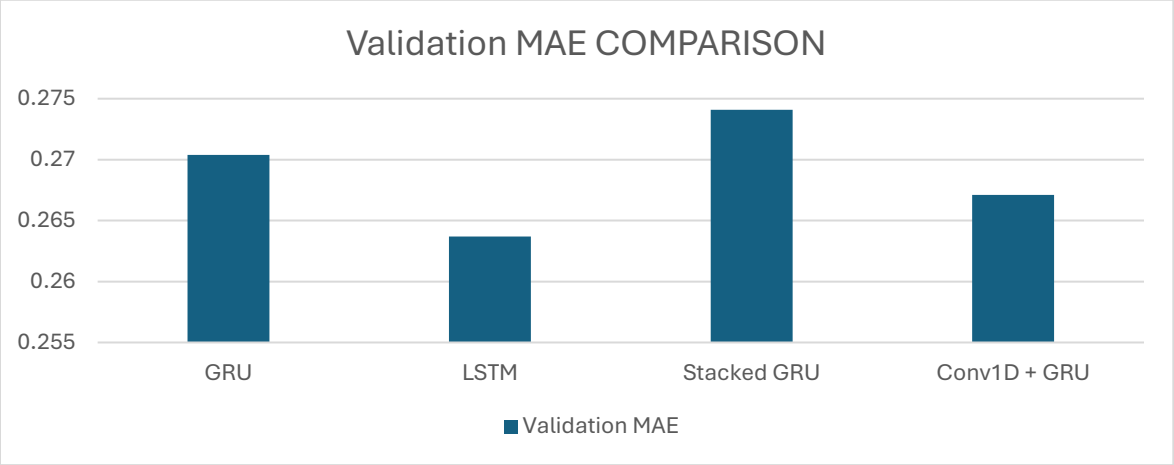


Figure 1. Validation MAE comparison of four deep learning models. LSTM achieved the lowest prediction error.

Based on these results, the LSTM model achieved the lowest validation MAE of 0.2637, making it the best-performing neural network in this experiment. The Conv1D + GRU model followed closely with an MAE of 0.2671, showing that combining convolutional and recurrent layers was also effective for temperature forecasting.

The basic GRU model achieved a validation MAE of 0.2704, while the stacked GRU model recorded 0.2741. Although the stacked model had additional recurrent layers, it did not improve performance compared with the simpler GRU architecture.

These findings suggest that increasing the complexity of a neural network does not automatically produce better predictions. In my experiment, the LSTM architecture was more effective at learning the temperature patterns, while the hybrid model also demonstrated competitive performance.

5. Test Set Evaluation

After comparing the validation results, I selected the two best-performing models, LSTM and Conv1D + GRU, for final evaluation on the test dataset. This step was important because the test data had not been used during model training or selection, allowing me to examine how well the models performed on unseen weather observations.

The final test results are summarized below.

Model	Test MAE (Normalized)	Test MAE (°C)	Test RMSE (°C)
LSTM	**0.2854**	**2.5028**	**3.2208**
Conv1D + GRU	**0.2882**	**2.5279**	**3.2342**

The LSTM model achieved the lowest test MAE of 2.5028°C, while the Conv1D + GRU model recorded 2.5279°C. This means that the LSTM predictions were approximately 2.50°C away from the actual temperatures on average.

I also compared the Root Mean Squared Error (RMSE), which gives greater weight to larger prediction errors. The LSTM model achieved an RMSE of 3.2208°C, slightly lower than the hybrid model's 3.2342°C.

Both models produced similar results, but the LSTM performed slightly better on the test dataset. This was consistent with the validation results, where LSTM also achieved the lowest MAE.

Overall, the test evaluation suggests that the LSTM model was the strongest forecasting approach among the architectures I tested, although the Conv1D + GRU model remained a competitive alternative.

6. Discussion and Final Conclusions

Through this assignment, I learned how different recurrent neural network architectures can be applied to weather forecasting and how their performance changes depending on the model structure. By comparing GRU, LSTM, stacked GRU, and Conv1D + GRU models, I was able to understand that selecting an appropriate architecture is more important than simply increasing the number of layers.

One interesting observation was that the stacked GRU model did not perform better than the simpler GRU model. I initially explored the deeper architecture to see whether additional recurrent layers would improve its ability to recognize weather patterns. However, the validation results showed that a more complex network does not necessarily lead to better forecasting accuracy.

The Conv1D + GRU model also produced encouraging results. Its validation MAE was close to that of the LSTM model, suggesting that combining convolutional and recurrent layers can be useful for identifying patterns in time-series data. However, the hybrid architecture did not outperform LSTM in this experiment.

The LSTM model achieved the best overall performance, with a validation MAE of 0.2637 and a test MAE of 2.5028°C. These results indicate that LSTM was effective at learning relationships between historical weather measurements and future temperatures.

Another important lesson from this experiment was the value of evaluating models on separate validation and test datasets. The validation results helped me identify the strongest models, while the test results provided a better understanding of their performance on unseen observations.

Overall, I concluded that **LSTM was the most suitable architecture among the models I tested for this weather forecasting task**. Although the other models were able to learn useful patterns, LSTM provided the lowest prediction errors. This assignment helped me gain practical experience in preparing time-series data, building recurrent neural networks, comparing model performance, and interpreting forecasting results.